

Open Source for Regulated Manufacturing

GMP-Compliant IIoT Data Logging with Apache IoTDB

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The moderne Pharmaceutical Manufacturing



Why Data Integrity Matters

- **Regulated Industries:** Crucial for life sciences, including Pharma, MedTech, and Biotech manufacturing.
- **The Core Conflict:** High-frequency IIoT industrial sensor data velocity colliding directly with strict compliance demands.
- **The Shift:** Evolving global frameworks require absolute transparency across the entire data lifecycle.
- **Motivation:** Can open source serve as a compliant historian backbone and pass a rigorous FDA/EU GMP inspection

The 2026 Regulatory Storm

- **The Hook:** Big Pharma is facing a massive regulatory shift heading into 2026[cite: 8, 9].
- **Annex 11 Overhaul:** Expanding from 1,500 to 10,000 words, bringing highly detailed requirements for data integrity, cloud computing, and IT security[cite: 9, 31, 162].
- **The New Annex 22:** Created specifically to regulate AI-supported and automated systems within GMP environments[cite: 10, 32, 163].
- **The Collision:** High-frequency IIoT devices are generating petabytes of data meeting the CPU/AI compliance era[cite: 10, 164].

The Regulatory Villain

- **Traditional Historians:** Proprietary ecosystems charging extortionate fees per data tag[cite: 11].
- **The Scaling Break:** Too expensive to scale; when handling high-frequency IIoT velocity, they break or bankrupt you[cite: 11, 12].
- **The Burning Question:** Can open-source software actually pass a rigorous FDA/EU GMP inspection better than proprietary options[cite: 13]?

 Graphic depicting AI/LLM restrictions and the introduction of Annex 22 in 2026

Compliance in 60 Seconds: FDA vs. EU (Part 1)

- **🔍 FDA 21 CFR Part 11 (US Regulation)** [cite: 39, 166]
 - Defines rules for electronic records & electronic signatures[cite: 40, 44].
 - Ensures electronic systems are trustworthy, secure, and legally equivalent to paper[cite: 41, 45].

Compliance in 60 Seconds: FDA vs. EU (Part 2)

- **EU Annex 11 (GMP Guideline)** [cite: 51, 167]
 - Governs computerized systems in GMP environments[cite: 52].
 - Covers the full system lifecycle including validation, risk management, and supplier governance[cite: 53, 54, 56].

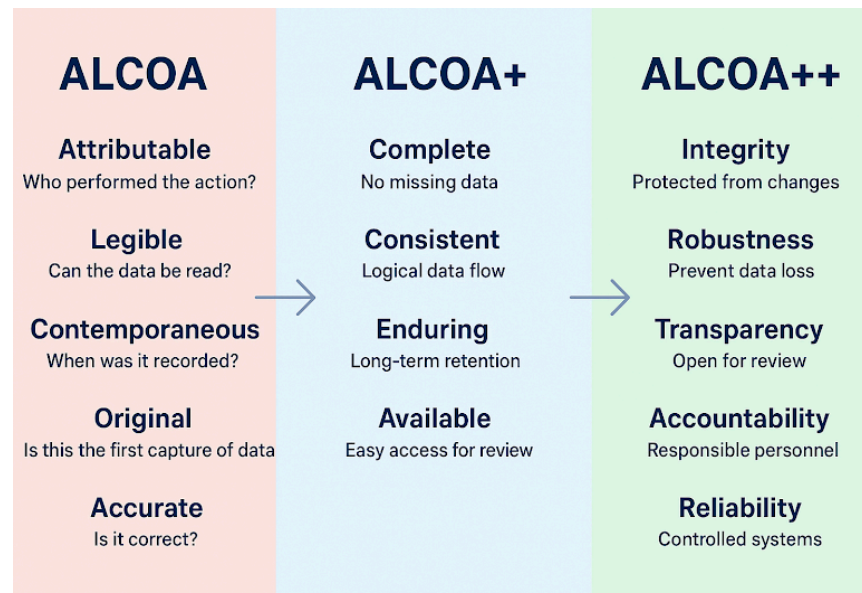
Compliance in 60 Seconds: The Takeaway

- **❓ Common Goal:** Ensure data integrity, traceability, and compliance[cite: 59].
- **The Digital Shift:** Enable audit-ready, reliable digital processes instead of paper[cite: 60].
- **❓ Takeaway:** Both regulations ensure digital GMP data is trustworthy across the full data lifecycle[cite: 65, 168].

GMP Requirements for IIoT Data

Demystifying ALCOA++ (The Core Framework)

- **The Gold Standard:** ALCOA+ is a framework ensuring the accuracy, consistency, and reliability of data in highly regulated industries[cite: 17, 72].
- **The Evolution:** Original ALCOA metrics expand into ALCOA+ and ALCOA++ to safely cover automated, high-velocity infrastructure[cite: 73, 174].



ALCOA: The Core 5 Principles

Principle	Definition
Attributable	Data must be clearly linked to the person or device that generated it [cite: 77].
Legible	Data must be readable, permanent, and accessible throughout its lifecycle [cite: 78].
Contemporaneous	Data must be recorded at the time of the activity, not retrospectively [cite: 79].
Original	Data must be the first capture of information, not a copy or transcription [cite: 80].
Accurate	Data must be correct, truthful, and free from errors or tampering [cite: 81].

ALCOA+: The 4 Extensions

Principle	Definition
Complete	All data, including deletions or modifications, must be retained and traceable [cite: 82].
Consistent	Data must be reliable, reproducible, and aligned with expected patterns [cite: 83].
Enduring	Data must be stored on durable media and remain accessible for required retention [cite: 84].
Available	Data must be readily accessible to authorized personnel when needed [cite: 85].

ALCOA++: The 5 Modern IIoT Safeguards

The Gap: Standard IT open-source tools fail ALCOA++ out of the box because mutable databases allow altering logs without audit trails [cite: 18].

The 5 Security & Telemetry Dimensions: [cite: 17, 86]

Principle	Definition
Integrity	Protected from unauthorized changes [cite: 86].
Robustness	Prevent data loss at high velocity [cite: 86].
Transparency	Open infrastructure design and review logic [cite: 86].
Accountability	Explicitly traceable to responsible personnel [cite: 86].
Reliability	Validated and controlled engineering environments [cite: 86].

Main Elements of 21 CFR Part 11

- **Validation:** Systems must be validated to perform exactly as intended[cite: 49, 175].
- **Security:** Access to electronic records restricted to authorized users[cite: 48, 176].
- **Record Retention:** Records must be securely retained in an unalterable environment[cite: 177].
- **Audit Trails:** Every action captured automatically with precise date/time stamps[cite: 47, 178].
- **Electronic Signatures:** Unique and indissolubly linked to their respective records[cite: 50, 179].
- **Training:** Personnel must be trained and accountable under Part 11 requirements[cite: 180].

 Infographic illustrating Validation, Security, Record Retention, Audit Trails, Electronic Signatures, and Training requirements

The Core Challenge

Why Traditional Historians Struggle (Commercial Constraints)

- **Proprietary Licensing:** Restrictive models based on strict data tag counts or ingestion volumes[cite: 11, 91, 181].
- **Vendor Lock-In:** Limited architecture flexibility and high vendor dependency[cite: 11, 92, 182].
- **Expensive Scaling:** Extreme costs to scale physical storage, high-availability, and retention[cite: 11, 93].
- **Validation Effort:** Exceptionally high validation and customization efforts inside a GAMP 5 framework[cite: 94, 183].

Why Traditional Historians Struggle (Technical Constraints)

- **Siloed Architectures:** Rigid data models offering minimal metadata, data lineage tracking, and query reproducibility[cite: 97].
- **Velocity Bottlenecks:** Inability to handle high-frequency sensor and rapid PLC time-series streams natively[cite: 69, 96].
- **Weak Auditing:** Demands intense layout customization to comply with automated ALCOA+ principles[cite: 98].
- **Integration Gaps:** Poor interface performance when talking to modern streaming analytics Engines[cite: 99].

The Industrial Data Reality: High-frequency sensor & PLC data needs rapid ingestion, massive capacity, and strict ALCOA+ audit trails simultaneously[cite: 100, 101, 102, 103].

The Role of Open Source

Open Source in Regulated Environments

- **The Transparency Advantage:** Closed-source platforms act as compliance "black boxes"[cite: 185]. Open source allows auditors to inspect data processing logic directly[cite: 105, 186].
- **Interoperability:** Open APIs and standardized file formats eliminate downstream analytical silos[cite: 105, 187].
- **Ecosystem Trust:** The Apache Software Foundation (ASF) provides community governance that minimizes vendor lock-in risks[cite: 106, 107, 188].

Open Infrastructure Data Commons (OIDC)

- **Real-World Baseline:** An open-source software platform leveraging Apache technologies for real-time monitoring and data preservation[cite: 109, 189].
- **UN Recognition:** Formally published within the United Nations STI Solutions Book, proving open infrastructure handles mission-critical, long-term data preservation[cite: 110, 117, 190].

 Screenshot of page 36 highlighting Open Infrastructure Data Commons using iotdb.apache.org

The Apache Solution

Apache IoTDB as a GMP Compliant IIoT Historian

- **Purpose-Built:** Optimized directly for high-frequency industrial sensors and PLC time-series streams[cite: 112, 113, 191].
- **Core Compliance features:** Structural mechanisms for unbending audit trail tracking, data versioning, and lineage[cite: 23, 114].
- **Security Controls:** Built-in user authentication, role-based access control (RBAC), encryption, and automated data retention management[cite: 115, 192].
- **UN Recognition:** Selected as one of the **Top 60 Global Innovation Cases** by the United Nations STI Forum[cite: 22, 117, 193].

Core Enabler: Apache TSFile

- **Immutable Design Pattern:** A columnar file layout that is append-only, serving as a regulatory dream because data cannot be silently modified or manipulated[cite: 21, 123, 194].
- **Data Segmentation:** Isolated physical data layout blocks enable clean validation boundaries and targeted spot-checks[cite: 123, 195].
- **Ecosystem Hybridity:** Combining TSFile with Apache Iceberg catalogs allows complex systems to track full historical query lineage and reproduce query states[cite: 124, 125, 126, 196].

Apache Projects for an End-to-End Pipeline

- **Data Integration & Edge:** `Apache PLC4X` parses OPC-UA/Modbus protocols natively[cite: 136, 197]. `Apache NiFi` handles automated data flow and integration[cite: 135, 198].
- **Messaging & Ingestion:** `Apache BifrostMQ` serves as a high-throughput MQTT broker [cite: 130, 199], feeding distributed `Apache Kafka` data streams[cite: 129, 199].
- **Stream Processing & Rules:** `Apache Flink` or `Apache Beam` execute real-time compliance filtering and threshold validation[cite: 131, 132, 200].
- **Storage & Analytics:** `Apache IoTDB` (hot storage) tracks time-series, while `Apache Iceberg` handles cold archives [cite: 143, 201][cite_start], analyzed via `Apache Superset`[cite: 139, 201].

Reference Architecture Map



End-to-end component diagram highlighting data paths from Edge Protocols through BifroMQ/Kafka up to IoTDB/TSFile and Iceberg Data Lakes

Compliance and Validation

Compliance Considerations

- **Annex 11 Mapping:** Matching specific clauses (e.g., electronic audit trails, system security, automated printouts) to configuration profiles in Apache IoTDB[cite: 146, 202].
- **21 CFR Part 11 Mapping:** Utilizing integrated user authorization matrices, electronic signatures concepts, and unalterable time-stamped logs[cite: 147, 203].
- **Validation Boundaries:** Clearly defining software boundaries between underlying operating systems, infrastructure, and containerized Apache applications[cite: 148, 204].

Computer System Validation with GAMP 5

- **The 9 Steps of Annex 11 Computer System Validation (CSV):** [cite: 88, 204]

1. Risk Assessment
2. Supplier Qualification
3. Requirements (URS)
4. Design & Specifications
5. IQ/OQ/PQ Testing
6. Data Integrity Controls
7. SOPs & Training
8. Change Control & Review
9. Operation & Monitoring



9-step GAMP 5 CSV linear map displaying milestones from Risk Assessment up to Operation & Monitoring

The 3-Tier Testing Blueprint (Step 5)

- **Installation Qualification (IQ):** Repeatable deployment checking using automated, reproducible container setups[cite: 148, 205].
- **Operational Qualification (OQ):** Verifying database failover operations, security profiles, user access, and automated data retention policies[cite: 148, 206].
- **Performance Qualification (PQ):** Simulating maximum industrial high-frequency IIoT velocity to guarantee stability and zero data loss under peak load[cite: 148, 207].

Conclusion

Key Takeaways

- **The Open Advantage:** Open-source ecosystems fully support transparent, scalable, and compliant industrial data platforms[cite: 151, 153, 208].
- **Future-Proof Storage:** Combining Apache IoTDB, TSFile, and Iceberg builds a modern, cost-effective, and audit-ready data lake landscape[cite: 152, 209].
- **Strategic Shift:** Adopting open source enables factories to navigate evolving regulations without getting trapped by legacy per-tag vendor licensing fees[cite: 210].

Q&A

- Thank you for attending!
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